	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 8

Lab Title: Understanding Decoders and Encoders

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Objective: To undesrating te basics of Encoder and Decoder experimentally verify the **.. j .** operation of Decoder and Encoder.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the Results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

EXPERIMENT 08

Decoder and Encoder

Objectives:

- To have a basic understanding of decoders and encoders.
- To experimentally verify the operation of decoder and encoder.

Equipment required:

- ICs as required
- Digital Electronics Trainer

Background Knowledge:

Decoder:

In digital electronics, a decoder can take the form of a multiple-input, multiple-output logic circuit that converts coded inputs into coded outputs, where the input and output codes are different e.g. n-to-2n , binary-coded decimal decoders. Decoding is necessary in applications such as data multiplexing, 7 segment display and memory address decoding.

Encoder:

An encoder is a device, circuit, transducer, software program, algorithm or person that converts information from one format or code to another.

Task 1:

Use basic gates to implement and design a 2 to 4 line decoder.

Steps:

1. Complete the truth table given below for a decoder
2. Write the logic equations formed.
3. Patch the equations on the trainer (do not unpatch the circuit, as it will be used later)

Truth Table					
A	**B**	**D3**	**D2**	**D1**	**D0**
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

EQUATIONS:

<table>
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<td> 

Y=A’B’</td>
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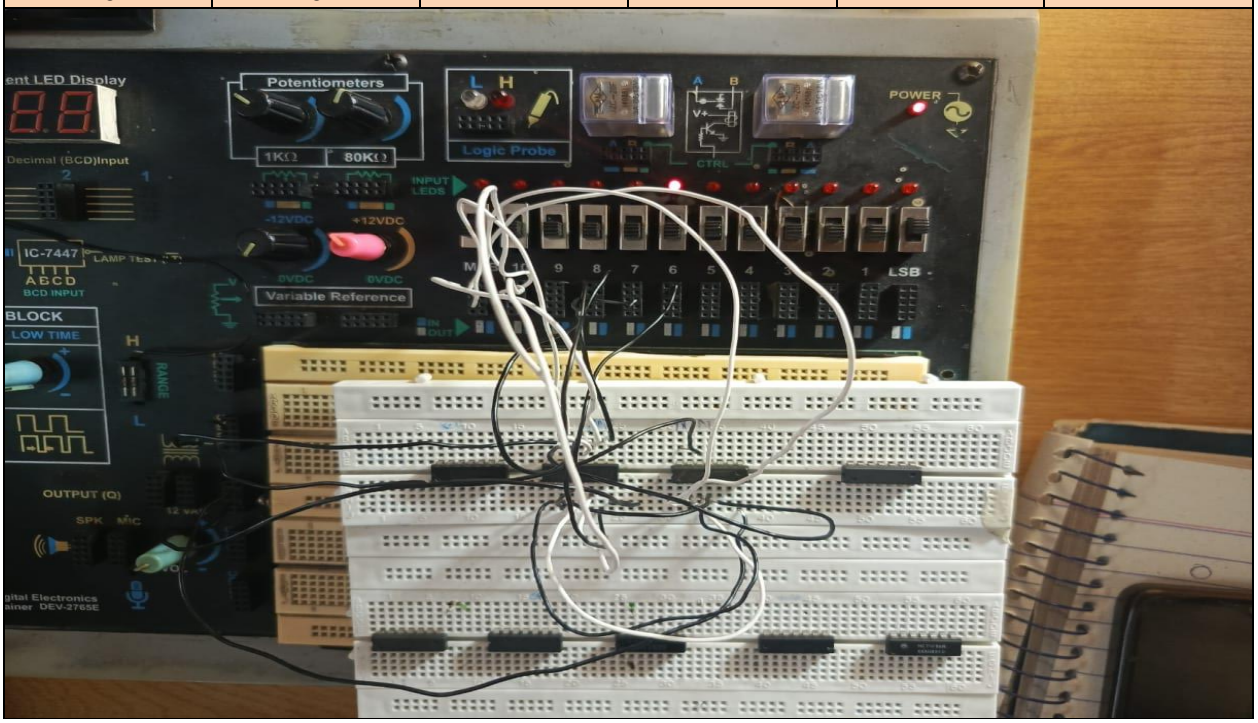
Y=A’B</td>
</tr>
<tr>
<td> 

Y=AB’</td>
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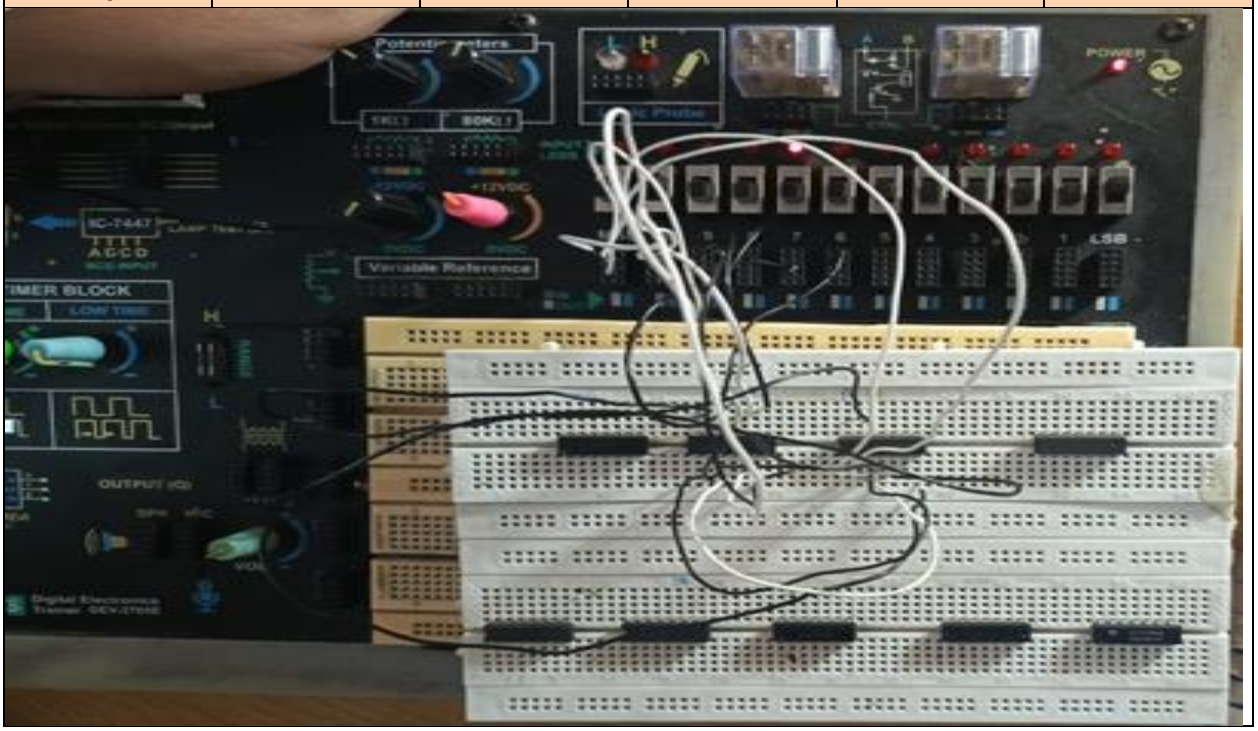
Y=AB</td>
</tr>
</table>

Task 2:
Simulation Results:

A	B	D3	D2	D1	D0
0	0	0	0	0	1



A	B	D3	D2	D1	D0
0	1	0	0	1	0

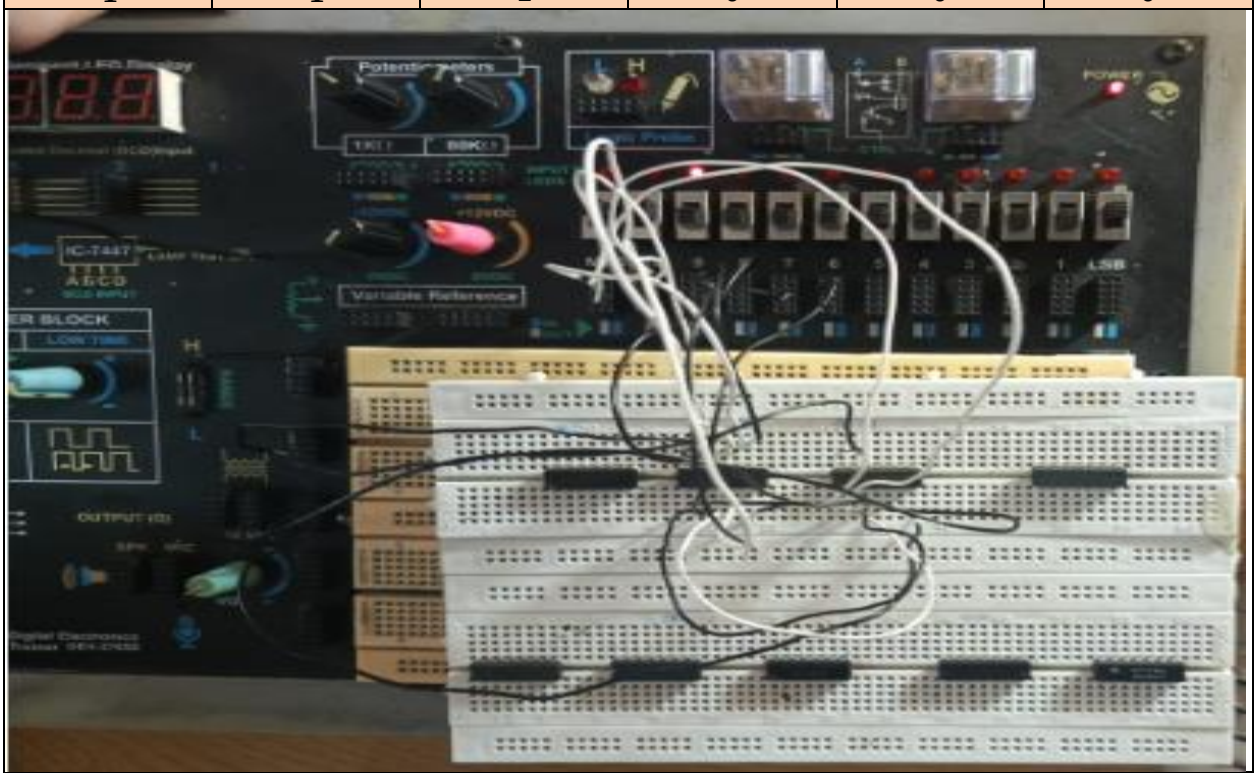


Task 2:

A	B	D_3	D_2	D_1	D_0
1	0	0	1	0	0



A	B	D_3	D_2	D_1	D_0
1	1	1	0	0	0



Task 2:
Use basic gates to implement and design a 4 to 2 line encoder.

Steps:

1. Complete the truth table given below for an encoder.
2. Write the logic equations formed.
3. Patch the equations on the trainer accordingly.

Truth Table					
D3	**D2**	**D1**	**D0**	**A**	**B**
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

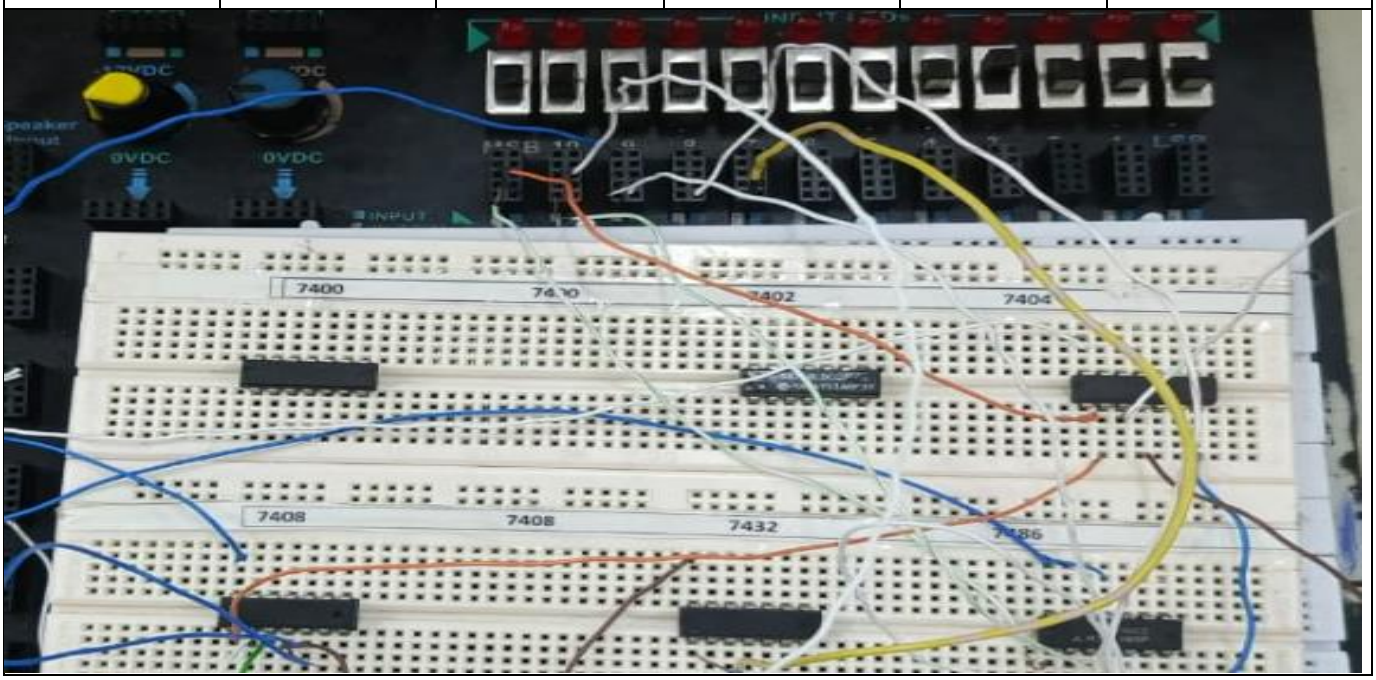
Equations:

From Truth table, we can write the Boolean functions for each output as:

$$A1 = Y3' Y2 Y1' Y0' + Y3 Y2' Y1' Y0'$$

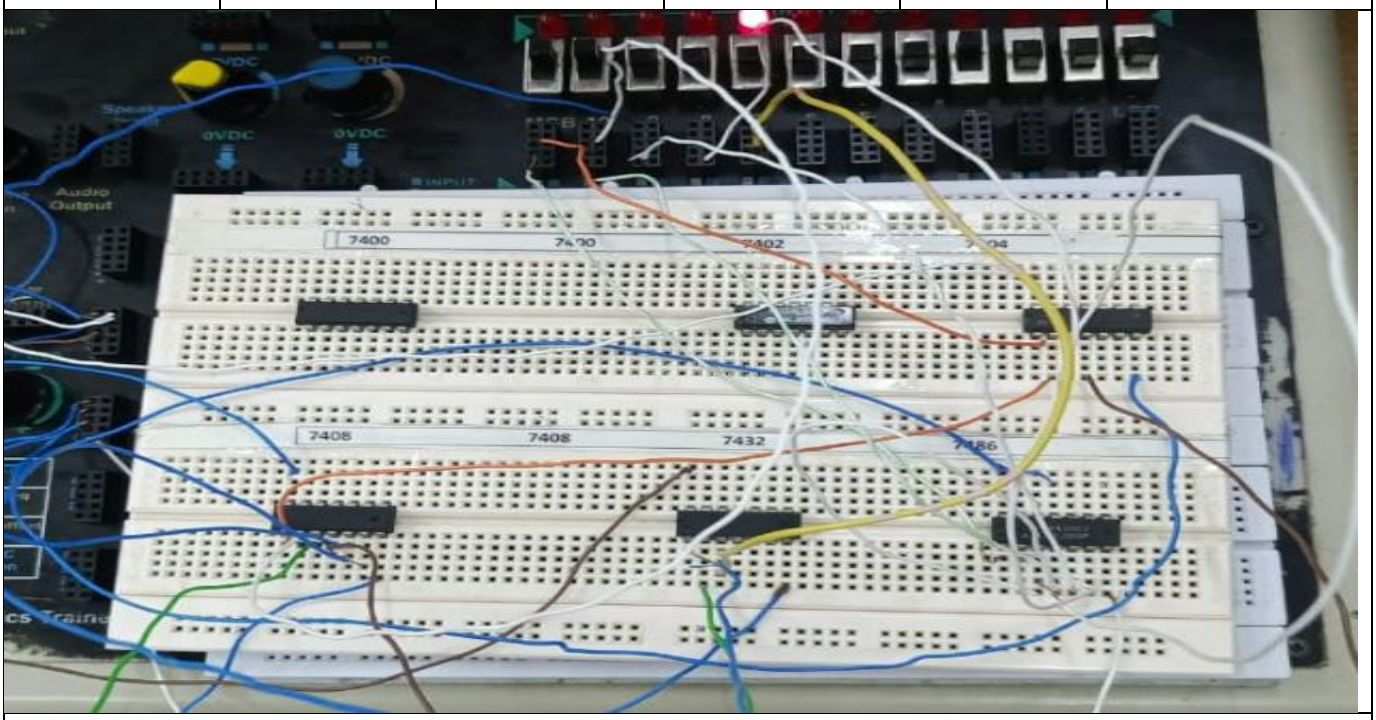
$$A0 = Y3' Y2' Y1 Y0' + Y3 Y2' Y1' Y0'$$

D3	D2	D1	D0	A	B
0	0	0	1	0	0

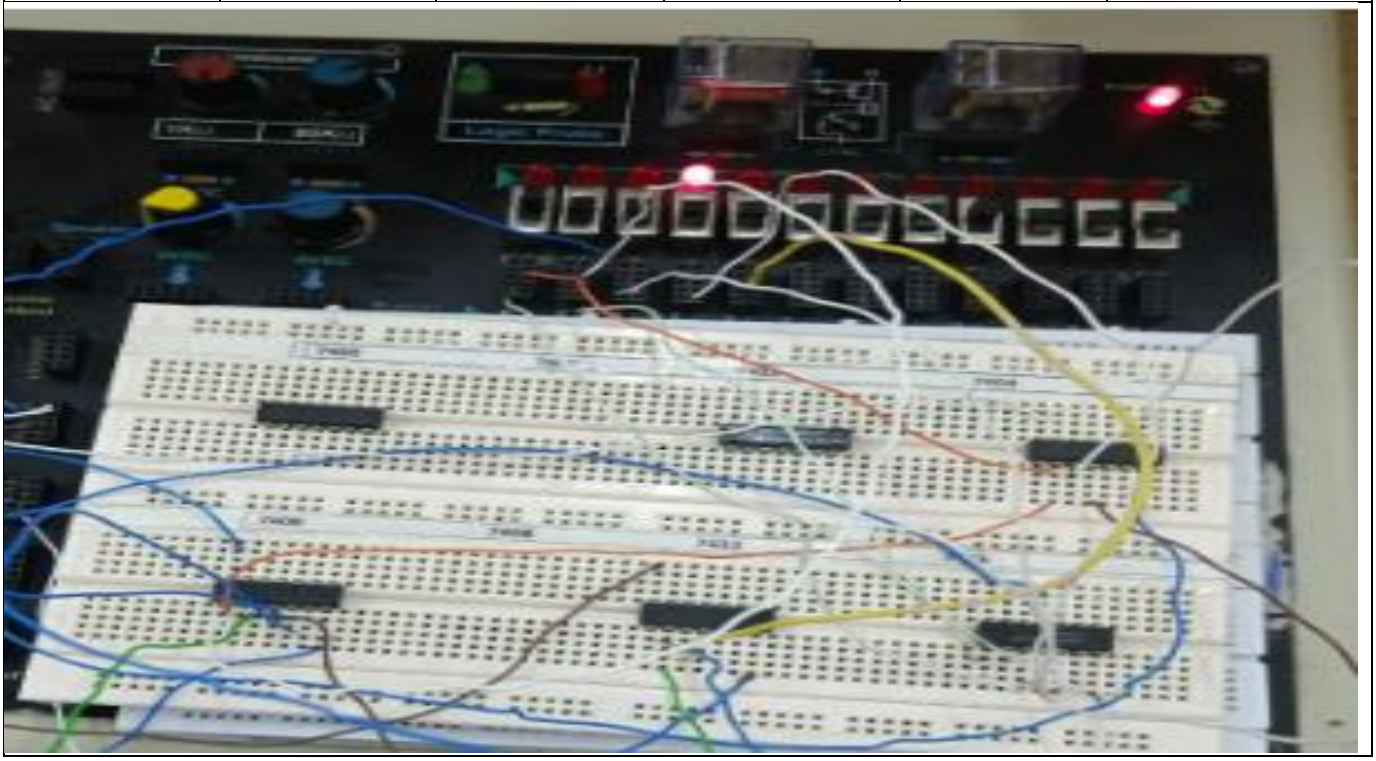


Simulation Results:

D3	D2	D1	D0	A	B
0	0	1	0	1	0

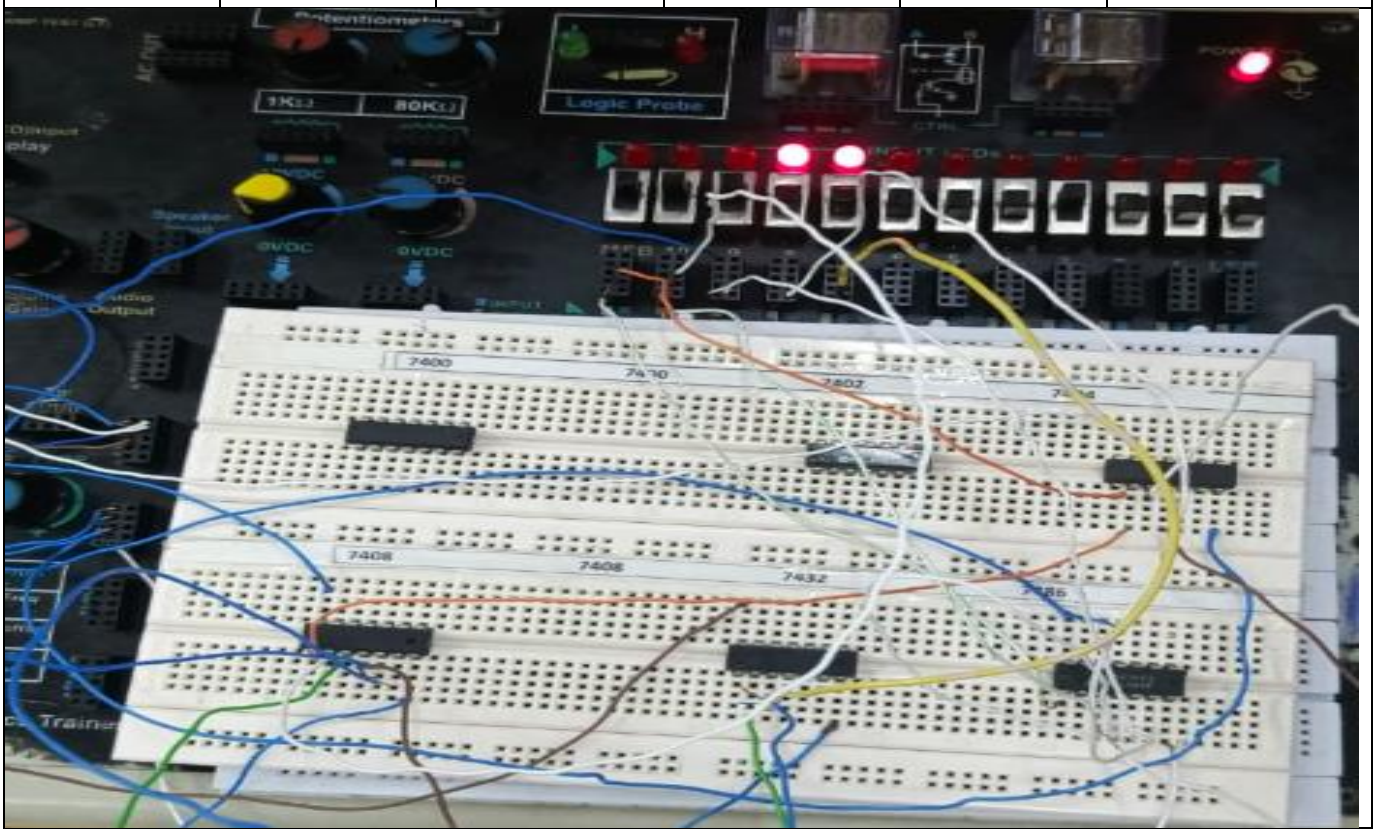


D3	D2	D1	D0	A	B
0	1	0	0	0	1



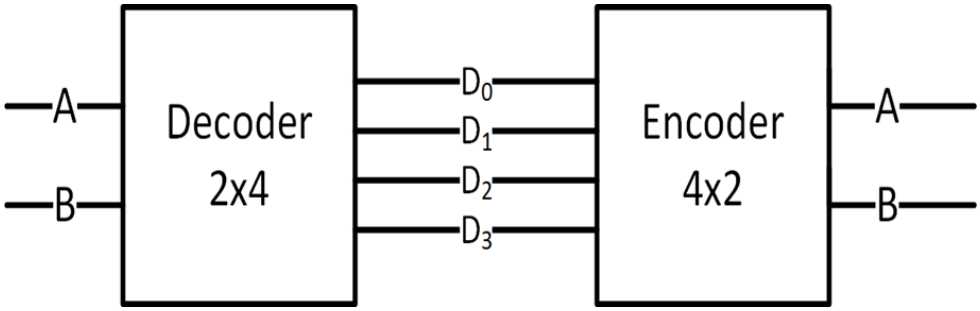
Simulation Results:

D3	D2	D1	D0	A	B
1	0	0	0	1	1



Simulation Results:
Task 3:

Consider the following diagram, which means if we connect a decoder and an encoder (of same order) in series we should get back our input.



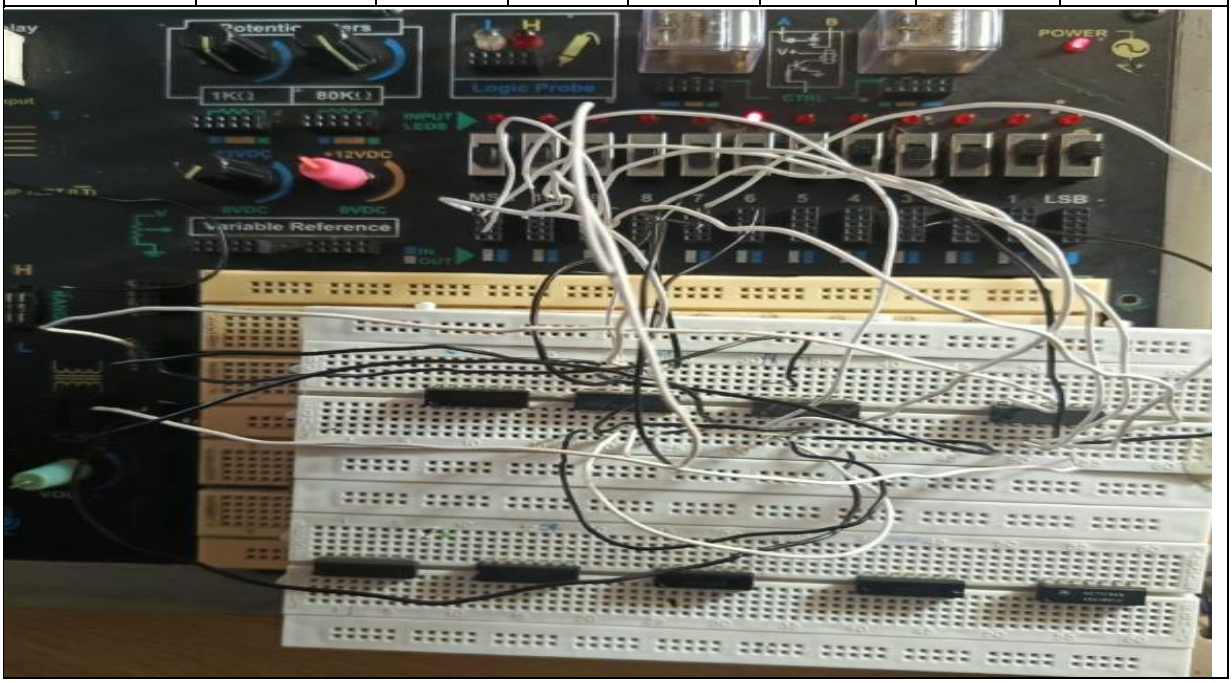
Connect the above designed decoder and encoder in series and verify the results.

Truth Table:

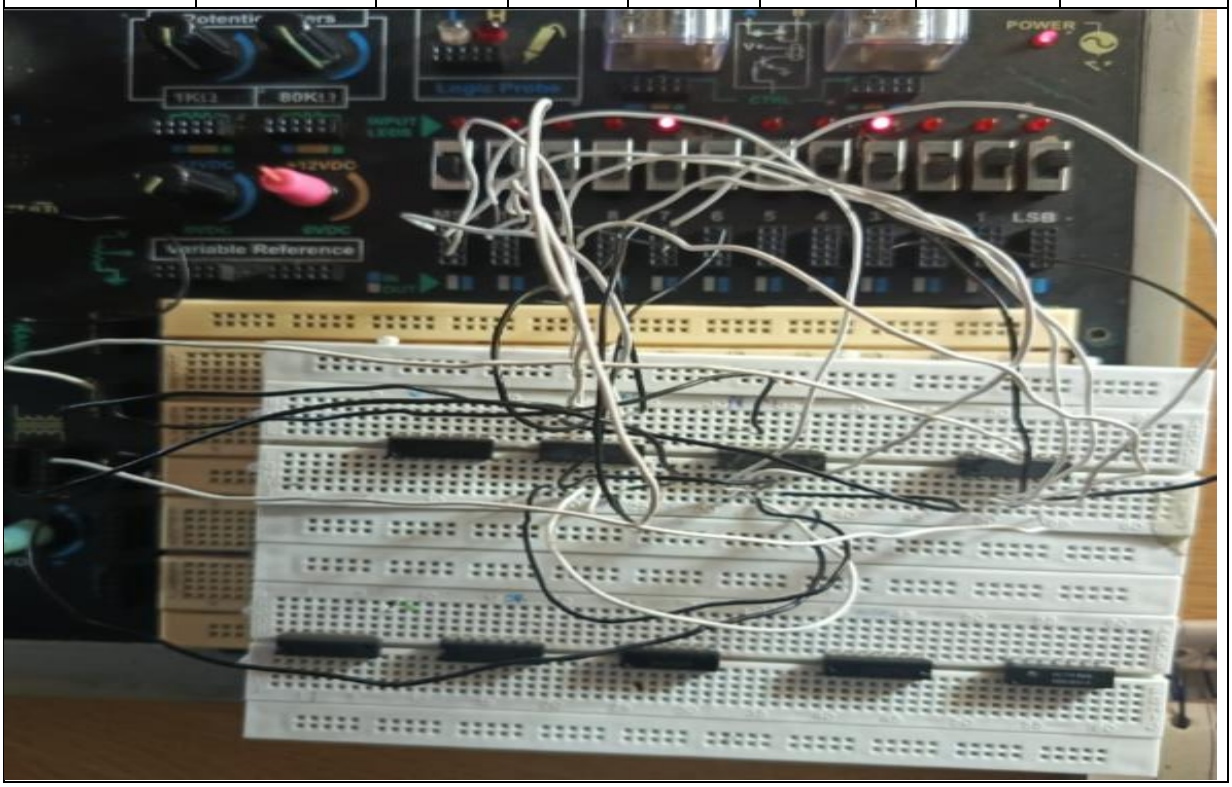
A	B	D0	D1	D2	D3	A	B
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	1
1	0	0	0	1	0	1	0
1	1	0	0	0	1	1	1

Simulation Results:

A	B	D0	D1	D2	D3	A	B
0	0	1	0	0	0	0	0

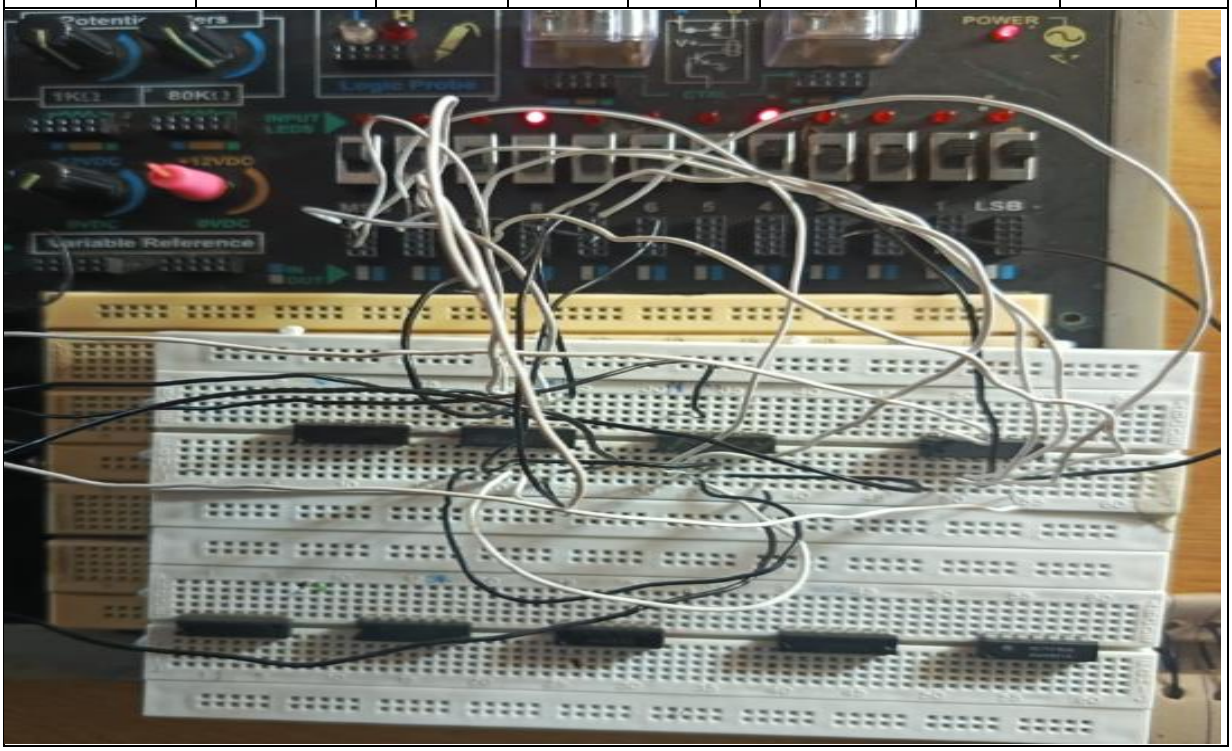


A	B	D0	D1	D2	D3	A	B
0	1	0	1	0	0	0	1

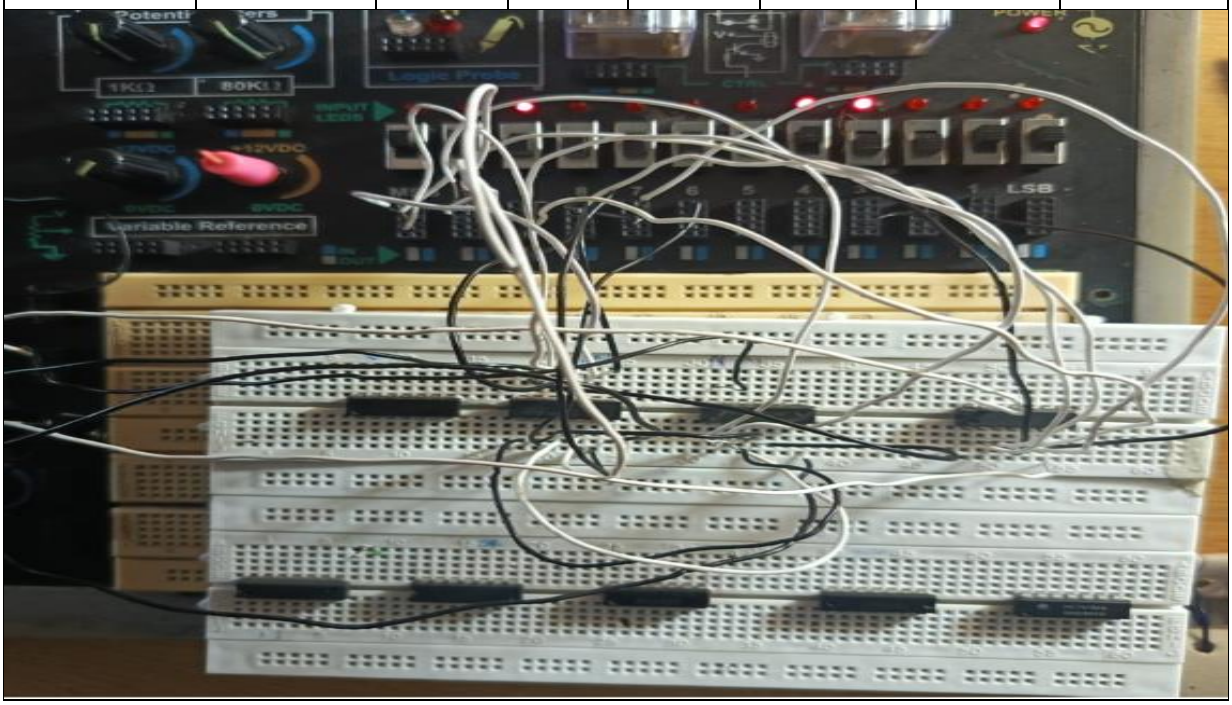


Simulation Results:

A	B	D0	D1	D2	D3	A	B
1	0	0	0	1	0	1	0



A	B	D0	D1	D2	D3	A	B
1	1	0	0	0	1	1	1



Simulation Results:

Conclusion:

In conclusion, understanding this setup shows the opposite-but-complementary relationship between an encoder and a decoder. The **decoder's** job is to take a short binary code (like `10`) and *expand* it into a single "one-hot" signal (activating only line 2 out of 4). The **encoder's** job is the exact reverse: it sees that single active line (line 2) and *compresses* it back into the original binary code (`10`). When connected, the decoder "unpacks" the input, and the encoder "re-packs" it, perfectly reversing the process. This demonstrates that they are inverse operations, which is why the final output is always an identical copy of the original input.